



### **Autumn Examinations 2014**

|                             |                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exam Code(s)</b>         | 1MF, 1SD, 1MDM                                                                                                                            |
| <b>Exam(s)</b>              | Masters in Software Design & Development<br>Higher Diploma in Applied Science (Software Design & Development)<br>Masters in Digital Media |
| <b>Module Code(s)</b>       | CT870                                                                                                                                     |
| <b>Module(s)</b>            | Internet Programming                                                                                                                      |
| <b>Paper No.</b>            | I                                                                                                                                         |
| <b>Repeat Paper</b>         | Yes                                                                                                                                       |
| <b>External Examiner(s)</b> | Prof. L. Maguire                                                                                                                          |
| <b>Internal Examiner(s)</b> | Prof. G. Lyons<br>Dr. M. Madden<br>* Dr. S. Redfern                                                                                       |

**Instructions:** Answer any three questions.  
All questions carry equal marks.

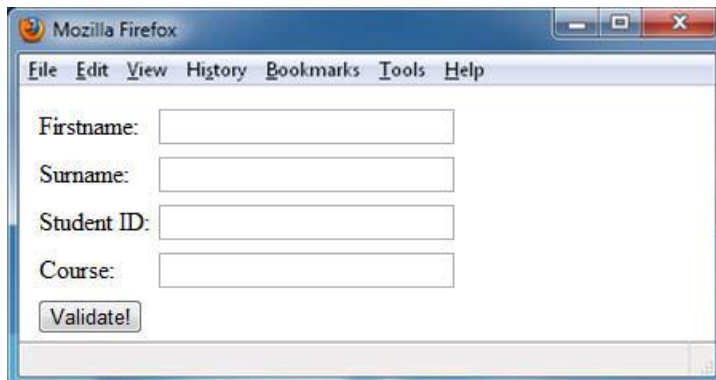
***Duration*** 2 hours  
**No. of Pages** 5  
**Department(s)** Information Technology  
**Course Co-ordinator(s)**

**Requirements:**

MCQ Release to Library: Yes ☐ No ☐  
Handout  
Statistical/ Log Tables  
Cambridge Tables  
Graph Paper  
Log Graph Paper  
Other Materials

### Q1.

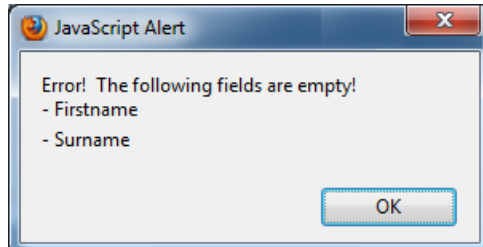
- a) With regard to HTML forms, what is the purpose of data validation? What are the advantages and disadvantages of server side and client side validation? Why might it be sensible to validate data twice, i.e. on both client and server? [6 marks]
- b) Write HTML code to produce the form as shown below.



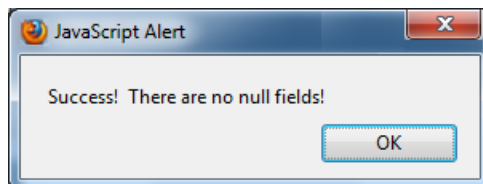
The screenshot shows a Mozilla Firefox browser window. The address bar is empty. The menu bar includes File, Edit, View, History, Bookmarks, Tools, and Help. The form contains four text input fields labeled 'Firstname:', 'Surname:', 'Student ID:', and 'Course:'. Below the fields is a button labeled 'Validate!'.

[3 marks]

When the Validate button is clicked, use client side validation to ensure that no fields are left null. If any fields are left null, an alert box similar to below should appear, informing the user which fields were left empty. (NB: It is not necessary to validate for data type, merely for null (blank) strings).



If all fields are completed, an alert box similar to the one shown below should appear



[5 marks]

- c) Write a simple webpage which uses Javascript to present a clock which updates once per second. The clock should be presented in this form:

The time is: **12:06:23**

[6 marks]

## Q2.

Create an HTML 'Contact Us' form. The form will accept a user's question details, and be posted to a PHP page which will insert the data into a MySQL database. Assume a MySQL database exists on the webserver, with a table called 'questions' containing the following fields:

| Fieldname | Datatype    | Description                                          |
|-----------|-------------|------------------------------------------------------|
| id        | int(10)     | A unique primary key for the question                |
| category  | varchar(50) | The category of question (goods, website, complaint) |
| name      | varchar(50) | The user's name                                      |
| email     | varchar(50) | The user's email address                             |

Write HTML code to produce a form similar to the one below. Note the use of a table to keep the form neat, and of colour styling. Note also that the drop-down (select) list contains three options: 'goods for sale', 'the website', and 'complaint' [6 marks]

|                   |                |
|-------------------|----------------|
| <b>Contact Us</b> |                |
| Question Category | Goods for Sale |
| Question          |                |
| Your Name         |                |
| Your Email        |                |
| <b>Submit</b>     |                |

Ensure that the 'contact us' form is validated server side so that the following conditions are met:

- No fields are null
- The email address supplied contains at least 6 characters, including exactly one '@' symbol and at least one '.' symbol [6 marks]

In the event of errors, messages should be displayed to the user highlighting the problematic field(s). Valid entries should be automatically filled into the text boxes so that they do not have to re-enter any fields that passed validation. [4 marks]

On successful validation of the form, the PHP script should insert the new details into the 'questions' table in the database. The user should also be presented with a message to confirm that the question was received successfully [4 marks]

### Q3.

- a) Write code for a CSS External Stylesheet (.css file) which defines the following styles:
- All <H1> elements should use 16 point sized text, coloured green
  - All elements whose class is 'blue' should use blue text
  - All other elements should use red text
  - All elements should use the Times New Roman font
  - All <th> elements should align their text centrally
  - All <div> elements have borders where the top border = 10 pixels, the bottom border = 5 pixels, the left border = 20 pixels, and the right border = 1 pixel in width

[6 marks]

- b) An online bookshop uses a MySQL database which contains a table called 'books' with the following schema:

| <u>Fieldname</u> | <u>Datatype</u> | <u>Description</u>                             |
|------------------|-----------------|------------------------------------------------|
| id               | int(10)         | A unique key for each book                     |
| title            | varchar(50)     | The title of the book                          |
| price            | float           | The price of the book (in Euro)                |
| quantity         | int(10)         | The stock level, i.e. quantity in the shop     |
| image            | varchar(50)     | The filename of an image of the book's artwork |
| author_id        | int(10)         | A foreign key to the <b>authors</b> table      |

The database also contains a table called 'authors', defined as follows:

| <u>Fieldname</u> | <u>Datatype</u> | <u>Description</u>                   |
|------------------|-----------------|--------------------------------------|
| id               | int(10)         | A unique key for each author         |
| name             | varchar(50)     | The author's name                    |
| country          | varchar(50)     | The country that the author lives in |

- Write PHP code to connect to the database and select all authors from the country 'Ireland' [3 marks]
- Extend this code so that the resulting dataset is displayed in an HTML page, with the details neatly laid out using a table. [4 marks]
- Write PHP code which receives a price (float) by query, (either by a POSTed form or a querystring). The PHP should display all book titles and author names where the book costs less than the specified price. You do not need to construct the HTML page which posts this data, and you do not need to display any book details other than title and author.

[7 marks]

Q4.

a) Write HTML code to produce the following table:

[4 marks]

| DC nationals |           |          |
|--------------|-----------|----------|
| Offense      | Defense   | Goalie   |
| Husmann      | O'Donnell | Weinberg |
| Popplewell   |           |          |
| McGilly      | Longo     |          |
| Donahue      | Seymour   |          |
| Camillo      | Walsh     |          |
|              |           |          |

b) Consider the following definition of a <div> tag:

```
<div id=sc style="width:800; height:587; position:absolute; left:0; top:0"> Hello </div>
```

- (i) Briefly explain what the 'id' attribute is for [2 marks]
- (ii) Briefly explain what each of the parameters (width, height, position, left, top) within the 'style' attribute are used for [4 marks]
- (iii) What is the role of the 'Hello' text in this code? [1 mark]

c) Write Javascript code to modify the content of the above <div> tag so that it displays on the browser the numbers 1 to 100, each separated by a line break.

[6 marks]

d) Explain, using code as appropriate, how you would provide a mechanism for the user to invoke the Javascript code you have just written.

[3 marks]